

AMAT113 Calculus II

Homework 2

Integration via Substitution and Integration by Parts

Submission Instructions

- Homework is due **Sunday at 11:59 PM Eastern Time**.
- Please print this PDF and complete the problems neatly by hand whenever possible.
- If printing is not available, completing the work on loose-leaf paper is acceptable.
- Show all necessary work. Answers without supporting work may receive reduced credit.
- Scan all completed pages, combine them into **one PDF file**, and submit the PDF through **Brightspace** before the deadline.
- Before submitting, confirm that every page is included, upright, legible, and in the correct order.

Part I: Integration via Substitution

1. Calculate each indefinite integral using the Substitution Rule. Write each final answer in terms of the original variable, not u , and include the constant of integration $+C$.

(a) $\int \sin^2 \theta \cos \theta \, d\theta$

(b) $\int 15x^2 \sqrt{x^3 + 1} \, dx$

(c) $\int \frac{1}{2\sqrt{x}(x+1)} \, dx$ Hint: Rewrite x as $(\sqrt{x})^2$.

(d) $\int \sec^2(3t - 1) \, dt$

Part I: Integration via Substitution (continued)

(e) $\int \frac{1}{x(\ln x)^2} dx$

(f) $\int x^2 e^{-x^3} dx$

(g) $\int \frac{\ln(\sin x)}{\tan x} dx$

(h) $\int \cos(x) e^{\sin x} dx$

2. Calculate each definite integral using the Substitution Rule. When rewriting in terms of u , change the bounds of integration appropriately.

(a) $\int_0^1 (3x - 1)^{50} dx$

(b) $\int_0^{\pi/6} \frac{\sin t}{\cos^2 t} dt$

(c) $\int_0^{\pi/3} \frac{\sin t - \cos t}{\sin t + \cos t} dt$

(d) $\int_0^1 \frac{y - 1}{y + 1} dy$

3. Calculate each integral. The answers involve inverse trigonometric functions.

(a) $\int \frac{1}{\sqrt{1-16x^2}} dx$

(b) $\int \frac{1}{25+16x^2} dx$

(c) $\int \frac{e^t}{1+e^{2t}} dt$

Part II: Integration by Parts

Integration by Parts Formulas

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$
$$\int_a^b f(x)g'(x) dx = f(x)g(x)|_a^b - \int_a^b f'(x)g(x) dx$$

4. Compute the following integrals.

(a) $\int x \cos(x) dx$

(b) $\int x^2 \ln(x) dx$

(c) $\int x^2 e^x dx$

(d) $\int e^x \cos(x) dx$

Part II: Integration by Parts (continued)

5. Compute the following integrals.

(a) $\int_0^{\sqrt{\pi}/2} x^3 \cos(x^2) dx$

(b) $\int_0^1 x(x+2)^8 dx$

(c) $\int_0^1 x^3 e^{2x} dx$

End of Homework 2